EFYL

Digital Address Book

Your address. Everywhere.

Complete User Guide

Version 2.1 — Sprint S4

This guide walks you step by step through **all** features of the EFYL Digital Address Book application, including offline mode, guided Household & Commerce surveys, the community map, QR code, data export and automatic synchronization.

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1. Product introduction

1.1 What is EFYL?

EFYL Digital Address Book is an **offline-first** mobile application designed to create, share and verify **accurate digital addresses** in Sub-Saharan Africa — and beyond — where classic street naming is missing, partial or outdated.

1.2 Why this application?

In hundreds of neighbourhoods, delivering a package, calling an ambulance, or simply indicating where one lives remains a headache. EFYL solves this problem in three motions:

1. **Create** a unique address by standing in front of one's door (e.g., `BES-AKW-3-12`) in less than a minute.
2. **Share** this address via WhatsApp, SMS, email, or a simple printable QR code.
3. **Verify** and collectively certify the position so it becomes reliable for everyone (couriers, emergency services, urban planning).

1.3 Who is this app for?

Profile	Main benefit
Households	Register their home and share easily via SMS or QR code.
Shop owners	Make their store findable by the whole neighbourhood on the community map.
NGOs & humanitarians	Census beneficiaries on the ground, even without a network.
City councils & postal services	Operate a digital reference registry.
Couriers & emergency services	Reach the right door on the first try.
Researchers & statisticians	Use geo-referenced data for socio-economic studies.

1.4 The 6 core product principles

Principle	What it means for you
Offline-first	Everything works without Internet. Network is a bonus, never a requirement.
Private by default	Your Household surveys remain strictly private. Only you see them.
Public commerces	Your Commerce surveys are voluntarily shared with the community.
GPS accuracy	Multi-sample measurement up to 1 m precision (sometimes less).
Community verification	Other users can confirm or flag your addresses.
Multilingual	French and English. Instant toggle.

2. First launch: onboarding & permissions

2.1 Welcome — Onboarding

On first launch, the app displays a **3-screen carousel** presenting:

- **Slide 1**: *“Create your address in an instant”* — GPS + door illustration.
- **Slide 2**: *“Share via QR Code, SMS, WhatsApp”* — sharing demo.
- **Slide 3**: *“Verify with the community”* — map with multiple pins.

At top-left, an **“FR / EN”** button lets you switch the language at any time. Your choice is remembered.

■ **Tip** — Tap **“Skip”** at top-right to jump to the next screen if you already know the app.

2.2 Device permissions

EFYL needs your permission to access some phone features. Requests are made **at the moment of use**, not all at once at startup:

Permission	When is it requested?	Required?
■ GPS Location	During the 1st address measurement (wizard step 1).	■ Yes
■ Camera	During 1st photo capture OR 1st QR scan.	■ Optional
■ Compass sensor	When locking entrance direction.	■ Optional
■ Storage	To download CSV/JSON exports.	■ As needed

■■ **Important** — If you deny GPS permission, the app will show an error and redirect you to the **system Settings** via a dedicated button. Once re-authorized, come back to EFYL and the measurement will resume.

2.3 Activate guest mode (no login)

At the end of onboarding, you have two choices:

- **Login by SMS** — receive a 6-digit code and get identified (useful for syncing on multiple phones).
- **Continue as guest** — everything works except your data is stored under the `guest` identifier. You can log in later without losing your data.

3. Authentication: SMS and guest mode

3.1 Login by SMS (OTP)

Authentication uses the **OTP** system (One Time Password — single-use code).

1. Enter your **phone number in international format**. Cameroon example: `+237 6XX XX XX XX`. The app also accepts `+33 6 12 34 56 78`, etc.
2. Tap **“Send code”**.
3. You receive a **6-digit code** by SMS (or shown on screen in demo mode).
4. Enter the code in the OTP field then **“Verify”**.
5. Once validated, you are redirected to the **Home** screen.

> ■■ **Current demo mode** — While Twilio is not enabled, the code **123456** works for **any phone number**. This info is shown on the login screen so you are never blocked.

3.2 Anti-spam protection (rate-limit)

To prevent an attacker from sending 1000 SMS in a loop, the app limits to **3 OTP code requests per phone per 10-minute window**. On the 4th attempt, you receive:

```
Error 429 — Too many requests. Please try again in 10 minutes.
```

3.3 Wrong OTP code

If you enter a wrong code, the app shows **“Invalid code”** and does not log you in. You can retry the code, or request a new code if you think you lost it.

3.4 Continue in guest mode

Guest mode is an express entry point to explore the app with no commitment:

- All create / survey / QR functions are available.
- Your data is saved **locally** on this phone, identified by `owner_id = guest`.
- When you decide to log in later, the data is **automatically migrated** to your real account (on the next synchronization).

3.5 Log out

From the **Profile** tab, tap **“Log out”**. Your data remains stored locally and on the server; you will simply be asked to log back in at the next launch.

4. Create a digital address: the 5-step wizard

This is **THE** core of the application. The wizard guides you step by step, with clear visual indicators (GPS pin color, locked satellites, progress bar).

Before you start

- **Physically stand in front of the entrance door** of the building to register.
- Do not cover the phone (step out from under a metal roof if possible — better signal).
- If possible, wait 10–20 seconds for satellites to lock before tapping “Confirm”.

Step 1 — Measure the door's GPS position ■

The app starts **sampling your position continuously**. Every second, it takes a new reading and computes a filtered weighted average (simplified Kalman) to smooth out occasional errors.

What you see on screen:

- A **precision circle** (green / yellow / red) based on signal quality.
- The **number of locked satellites** (typically 4 to 12 — the higher the better).
- **Latitude / longitude** coordinates in decimal degrees updated in real time.
- **Accuracy in meters** (desired ≤ 1 m, acceptable up to 5 m).

GPS pin color code:

Color	Accuracy	Advice
■ Green	≤ 1 m	Excellent! You can confirm.
■ Yellow	1 to 5 m	Acceptable. Wait a bit more to improve.
■ Red	> 5 m	Poor. Move a few meters or wait.
■ Grey	Unknown	No signal yet — wait 10 to 20 s.

When should I tap “Confirm GPS Position”?

Ideally when the pin is **green** and stable for 5–10 seconds. If accuracy stalls at yellow (e.g., indoors, under dense trees, or in narrow alleys), a **“Save anyway”** button appears after 30 seconds.

> ■■ **Warning** — If you save with accuracy > 5 m, the address will be marked **“Low accuracy”** with a red badge. You (or another user) can correct it later via the **“Verify address”** function.

Step 2 — Address details ■

Once the position is confirmed, the app automatically queries the **Nominatim (OpenStreetMap)** service to **auto-detect the district**. If you are offline, this field stays empty — you'll fill it in manually.

Fields to fill in:

Field	Description
District	Code prefix (e.g., BES for Bessengue). Auto-detected.
Sector	Internal zone code (e.g., AKW). You enter it.
Street number	Street identifier (digit).
House number	Your door number on the street.
Contact name	Your name or owner's name (optional).
Phone	Number to reach you (optional).
Notes	Any useful detail: “red door across from the pharmacy”.

■ **Unique ID construction** — The final address ID is automatically generated as `DISTRICT-SECTOR-STREET-HOUSE`, example: `BES-AKW-3-12`. This is the ID you will share.

Step 3 — Entrance direction & photo ■■

This **optional** but very useful step captures two key pieces of information:

3.A — Azimuth (compass)

1. Hold the phone flat, screen facing up.
2. **Point the top of the phone toward the entrance door** of the building.
3. An animated compass appears, with an angle in degrees (0° = North, 90° = East, 180° = South, 270° = West).
4. Tap **“Lock direction”**. The angle is saved.

This azimuth will be shown to future visitors as: *“The entrance faces North-East (45°)”*.

3.B — Building photo

1. Tap **“Take a photo”**.
2. Camera opens. Frame the building façade or door.
3. Validate. The photo is encoded in **base64** and stored locally in the address record.
4. You can “Retake” if the photo is not satisfactory.

Step 4 — Manual satellite correction ■■

Sometimes, even with good GPS accuracy, the point can be slightly off (a few meters). This step lets you **manually correct** the position on a **satellite image**.

1. The map opens centered on the measured GPS position, with a draggable orange pin.
2. Switch to **“Satellite”** mode via the top-right button if needed.
3. Zoom (pinch) to clearly see your roof or yard.
4. **Drag the orange pin** exactly to the desired spot.
5. Tap **“Validate correction”**.

> ■ **Data preserved** — The raw GPS position (`lat_raw / lng_raw`) and corrected position (`lat_corrected / lng_corrected`) are stored separately. Nobody ever loses the original raw data.

Step 5 — Final review and save ■

The summary screen shows **all** the information entered:

- GPS position (raw and corrected if applicable).
- Measured accuracy + badge.
- Generated address ID.
- District / sector / street / number.
- Direction (azimuth).
- Photo (thumbnail).

You can **go back** on each step via the top tabs. When everything is correct, tap **“Save”**.

> ■ **What happens next?**
> 1. The address is saved to local storage **instantly**.
> 2. It appears on the **Home** screen.
> 3. A **“CREATE”** operation is added to the sync queue.
> 4. If you are online, sync triggers automatically.
> 5. The app immediately offers to **start the survey** (Household or Commerce).

5. Guided surveys: Household or Commerce?

Right after the address is created, the app offers you to fill in a **survey**. This survey enriches your record with socio-economic (Household) or commercial (Commerce) information. It is **optional** but highly recommended — this is what makes your address truly useful for the community or statistics.

5.1 The first question: Household or Commerce?

You see **two big cards** side by side:

■ Household	■ Commerce
Home / dwelling. Data is private, visible only by you.	Shop / business. Data is public, visible to everyone. 19 steps (people, equi

■ **Privacy critical** — This distinction is crucial. If you choose “Household”, **NO ONE** else will ever see these details. If you choose “Commerce”, your shop will be **visible to everyone** on the community map, just like a shop on Google Maps.

5.2 Household path — 19 steps

If you choose **Household**, the wizard takes you through the following steps:

Step	Description
1. Total persons	Total household member count (e.g., 5).
2. Main income source	Free text: “Commerce”, “Agriculture”, “Civil service”...
3. Other income sources	Free text, comma-separated.
4. Income level	Choice: Very low / Low / Medium / High.
5. Phones	Number of phones in the household (smartphones + classic).
6. Internet	Yes / No — Do you have Internet at home?
7. Electricity	Stable / Unstable / None.
8. Water	Stable / Unstable / None.
9. Toilets	Flush / Traditional / Other (text field if Other).
10. Kitchen	Position: In a room / In the living room / Separate.
11. Living rooms	Total number of living rooms.
12. Floors	Number of floors of the building.

13. Walls	Concrete / Adobe / Wood / Other.
14. Glass windows	Yes / No.
15. Roof	Metal / Wood / Plastic / Concrete / Other.
16. Vehicles	Sub-questions: bicycles, motorcycles, cars, trucks.
17. Persons intro	Intro screen before each person entry.
18. Person record (repeated)	Name, phone or email, age, gender, education, profession, frequented place.
19. Summary	Full review and “Save” button.

■ **Tip** — You can add as many persons as you want. Tap **“+ Add a person”** after each record. When done, **“Finish entry”** takes you to the summary.

5.3 Commerce path — 3 steps

If you choose **Commerce**, the wizard is much shorter — just fill in the info useful to future customers:

Step	Fields to fill
1. Household/Commerce choice	(Already done — you chose Commerce)
2. Commerce information	Business name (required), Owner, Phone, Products/activi
3. Summary	Review + “Save” button.

5.4 What happens to the survey after saving?

1. It is stored **locally** in AsyncStorage (`surveys` key).
2. An **UPSERT** operation is added to the sync queue (`survey_queue` key).
3. The linked address is marked `has_survey=true`.
4. If you chose Commerce, the address is also marked `is_commerce=true` and becomes **public automatically** on the next sync.
5. On the address card, a ■ **Commerce badge** appears.

6. Interactive map and community map

6.1 Open the map

Tap the **Map** tab at the bottom of the screen. The map (based on Leaflet + OpenStreetMap) opens, automatically centered on all your addresses.

6.2 Understanding pins

Pin color	Meaning	Clickable?
■ Green	Verified address by community.	■ Yes — opens the record.
■ Yellow	Address created but not yet verified.	■ Yes.
■ Red	Address flagged 3+ times (suspicious).	■ Yes.
■ Grey	Address with low accuracy (> 5 m).	■ Yes.
■ Orange	Community commerce (shared by another).	■ Yes.

6.3 “Community commerces” toggle

Just below the map header, a button with the ■ icon shows / hides **commerces shared by all other users**. By default, this toggle is **ON** (you see community commerces).

> ■ **Good to know** — When the toggle is ON, you see both **your** addresses (households + commerces) **and** commerces shared by others. When OFF, only your addresses appear. Other users' households are **NEVER** visible, regardless of the toggle.

6.4 Switch Street ■ Satellite

At the top right of the map, two buttons:

- **“Streets”** — classic OpenStreetMap display (drawn streets, buildings).
- **“Satellite”** — high-resolution aerial image (Esri World Imagery).

Satellite mode is very useful to visually verify the building and correct the position if needed.

6.5 Zoom, pan, navigate

1. **Zoom**: pinch with two fingers (in) or spread two fingers (out).
2. **Pan**: drag with one finger.
3. **Recenter on my location**: target icon bottom right.
4. **Tap a pin**: opens a popup with name and ID, and a “View record” button.

6.6 Offline behavior of the map

■■ **Current limitation** — The Leaflet map downloads **tiles** (small images) from OpenStreetMap at display time. Offline, only tiles already consulted recently remain visible. Uncached zones appear white. The **pins (your data)** remain perfectly displayed.

7. QR Code & address sharing

7.1 Generate an address QR Code

1. From the **Home** screen, tap an address to open its record.
2. Tap the **“QR Code”** button at the top right.
3. A large QR Code appears on screen, along with the address ID in plain text.

The QR contains a special URL `efyl://address/BES-AKW-3-12` that, when scanned by any phone with EFYL installed, directly opens the corresponding address record.

7.2 Share via WhatsApp / SMS / Email

Below the QR Code, several share buttons:

Button	Behavior
■ **WhatsApp**	Opens WhatsApp with a prefilled message containing the address ID and link.
■ **SMS**	Opens the SMS app with the prefilled message.
■ **Email**	Opens your email client with prefilled subject and body.
■ **Generic share**	Shows the native system share menu (Bluetooth, AirDrop, etc.).

7.3 Scan another user's QR Code

1. Go to the **Scan** tab (camera icon).
2. Grant camera access if asked.
3. Point the camera at the QR Code to scan.
4. The app automatically detects the QR and displays the address information.
5. Three options appear: **“Add to my book”**, **“View directions”**, **“Ignore”**.

■ **Directions** — The “View directions” button opens your favorite navigation app (Google Maps, Waze, Apple Maps...) with the destination already entered at the GPS coordinates.

7.4 Print a QR Code

You can also **print** the QR Code on a sticker to put on your door. On the QR screen, the **“Download PNG”** button saves the image to your gallery, which you can then print from any internet café or family printer.

8. Community verification

8.1 Why verify?

The cross-verification system is what makes EFYL **reliable in the long run**. If someone creates an approximate or false address, other users can correct it, and the address gets **certified** or **flagged**.

8.2 How to verify an address

1. **Physically** go to the address to verify (or very close to it).
2. Open the address record (from your book or by scanning someone's QR).
3. Tap **“Verify this address”**.
4. The app measures your current GPS position.
5. It computes the **haversine distance** between your position and the saved one.
6. Tap **“Submit”**.

8.3 What happens next?

Measured distance	Result
≤ **10 meters**	Address VERIFIED ✓ Pin turns green. App shows: <i>“Position v
> 10 meters	Address flagged The flag_count counter is inc
After 3 flags	Address marked SUSPICIOUS Pin turns red. The owner can ther

■ **System spirit** — Verification is **voluntary and benevolent**. It does not replace the address creator, it **complements** their trust. Each verification strengthens or nuances the overall reliability of the database.

9. The OFFLINE mode explained in depth

■ **Essential section** — Offline mode is EFYL's **signature**. This section explains step by step how it works technically, so you understand what happens under the hood.

9.1 The core principle: Offline-First

EFYL is built on the **Offline-First** paradigm: network connectivity is an **optional bonus**, never a condition of operation. All essential operations — create, edit, delete, photograph, share — first go through local storage, then are **deferred-synced** with the server as soon as the network is available again.

9.2 GPS works WITHOUT Internet

This is a very common misconception: **GPS does NOT need Internet**. Here's why:

- Your phone's GPS directly receives signals from satellites in orbit (**American GPS**, **Russian GLONASS**, **European Galileo**, **Chinese BeiDou**).
- These satellites broadcast **continuously**, and your phone captures their signals via antenna.
- The phone itself computes your position via **triangulation** (distance × time).
- **Result**: even deep in the Cameroonian bush or in a desert, without any cellular signal, your position is known to within 1–5 meters.

■■ **Tech detail** — “Assisted GPS” (A-GPS) uses the cellular network to pre-fetch satellite ephemerides faster. Without a network, the first fix can take 30–60 seconds instead of 5–10. But once acquired, everything works identically.

9.3 Local storage: AsyncStorage

All your data is stored in **AsyncStorage**, the persistent key/value storage system provided by React Native (mobile equivalent of localStorage). Here's the structure of keys used:

AsyncStorage key	Contents
addresses	JSON list of all your created addresses.
surveys	JSON list of all surveys (Household + Commerce).
sync_queue	Queue of CREATE / DELETE operations to sync.
survey_queue	Queue of UPSERT survey operations to sync.
last_sync	Timestamp of the last successful synchronization.
prefs	User preferences (language, theme, etc.).
user	ID and phone of the logged-in user.

Persistence: this data survives app closure, phone reboot, and even battery death. Only an app uninstall erases it.

9.4 The lifecycle of an offline-created address

```
1. You tap 'Save'
↓
2. App stores the address in AsyncStorage ('addresses' key)
↓
3. App adds an operation to 'sync_queue'
Format: [{action: 'CREATE', payload: {...}, ts: ...}]
↓
4. IMMEDIATE confirmation to user ('Address created!')
↓
5. User can keep creating more addresses
for days without network
↓
6. When network returns (NetInfo detection):
a. processSyncQueue() is triggered
b. Each operation is replayed to the backend
c. Backend confirms or returns an error
d. Successful operations are removed from the queue
e. Failed operations stay for the next attempt
```

9.5 Automatic network detection (NetInfo)

We use the `@react-native-community/netinfo` library that continuously listens for network state. Here's the simplified code:

```
NetInfo.addEventListener((state) => {
if (state.isConnected && !wasConnected) {
// Network just came back!
processSyncQueue(); // Drain the queue
refreshAddresses(); // Refresh from server
refreshPublicCommerces();
}
wasConnected = state.isConnected;
});
```

As soon as your phone picks up WiFi, 4G or 3G, **within the second that follows**, the queue automatically drains without any action on your part.

9.6 Reverse geocoding: the only exception

■■ **ATTENTION — Reverse geocoding** (auto-detecting district from GPS coordinates) is **the only feature** that needs Internet, since it calls the OpenStreetMap Nominatim API.

Offline consequence: the “District” field remains empty. You fill it in manually and the address is created normally.

9.7 Online vs Offline recap table

Feature	Offline?	Notes
■ Measure GPS position	■ YES	Satellites, no Internet.
■ Create an address	■ YES	Instant AsyncStorage.
■ Household/Commerce survey	■ YES	Instant AsyncStorage.
■ Take building photo	■ YES	Local camera.
■ Compass (azimuth)	■ YES	Phone's local sensor.
■ Generate QR code	■ YES	100% local computation.
■ Share via WhatsApp/SMS	■■ Partial	Opens the app — sends when network returns.
■■ District auto-detection	■ NO	Needs Nominatim.
■■ New map tiles	■ NO	Needs OSM.
■■ Backend sync	■ NO	But auto on network return.
■ SMS login (Twilio)	■ NO	But guest mode available.
■ Scan another user's QR	■ YES	100% local decoding.
■ See community commerces	■ NO	Needs backend fetch.
■ Verify an address	■ NO	Needs to send to backend.
■ CSV/JSON export	■ NO	Backend endpoint.

9.8 What to do if blocked offline?

- **District is not auto-detected** → fill it in manually.
- **Map is white** → position is still measured, create the address, you'll see the map when network returns.
- **I can't see community commerces** → normal, wait for network return.
- **I can't verify an address** → also normal, do it when network returns.
- **Sync queue counter grows** → this is **normal**, it just means you have pending operations. Everything will be sent automatically when network returns.

10. Automatic and manual synchronization

10.1 The connectivity indicator

At the top of the Home screen:

- **No indicator** → you are online, everything is synced.
- **Yellow "Offline" banner** → you are offline, but data is saved locally.
- **"X pending" badge** → you have X operations waiting to sync.

10.2 Auto-sync on network return

As soon as NetInfo detects connection return (WiFi or cellular), sync triggers without any action. You will see:

1. The yellow **"Offline"** banner disappears.
2. A discreet **"Syncing…"** notification at the bottom.
3. The pending operations counter drops to 0.
4. The **"Last sync"** timestamp in Profile is updated.

10.3 Manual sync

If you want to force sync (for example, you know a teammate just created commerces and you want to see them):

1. Go to the **Profile** tab.
2. Tap **"Sync now"**.
3. Wait a few seconds — a progress indicator shows.
4. Confirmation notification at the end.

10.4 What exactly gets synced?

Direction	Data concerned
Phone → Server	Addresses created, edited, deleted offline. Household/Commerce surveys. Of
Server → Phone	Recent public commerces. Verifications done by other users. Status updates

10.5 Conflicts and handling

If you edit an address offline while another user edits it online, the system applies the **“Last Write Wins”** rule (the latest timestamped change wins). For critical addresses (verification), the backend always favors the **most recent** data.

11. CSV / JSON export of your data

11.1 Why export?

- Make a local backup on your computer.
- Analyze your data in Excel or LibreOffice.
- Share a list of addresses with a partner.
- Provide data to a postal service, NGO or city council.

11.2 How to export

1. Go to the **Profile** tab.
2. “My data” section.
3. Tap **“Export as CSV”** or **“Export as JSON”**.
4. The file is downloaded into your phone's **Downloads** folder.
5. You can share it (WhatsApp, email, AirDrop, Google Drive...) or open it directly.

11.3 What's in the CSV file?

The CSV file contains one row per address, with the following columns (and more):

```
address_id, latitude, longitude, accuracy_meters, lat_raw, lng_raw,
lat_corrected, lng_corrected, azimuth_degrees, photo_base64,
is_commerce, has_survey, owner_id, district, sector, street_id,
house_number, contact_name, contact_phone, notes,
verified, flag_count, low_accuracy, created_at,
survey_total_persons, survey_main_income_source, survey_electricity,
survey_water, survey_walls, survey_roof, ...
commerce_name, commerce_owner, commerce_phone, commerce_products,
commerce_hours_weekdays, ...
```

11.4 JSON format for integrations

The JSON format is more structured and allows nesting surveys and the persons list. Ideal for developers or integrations with other systems (API, external database...).

12. Profile & settings

12.1 Access the profile

Profile tab at the bottom right. You will find:

- Your **phone number** (or “Guest mode”).
- Your **personal statistics**: count of addresses, surveys, commerces, verifications done.
- Status of **last sync**.
- **Settings**: language, theme, etc.
- **Actions**: export, sync, log out, delete data.

12.2 Change language

1. Profile → **“Language”**.
2. Choose **Français** or **English**.
3. The whole app switches **instantly** without restart.

12.3 Delete all my data

> ■ **IRREVERSIBLE ACTION** — This function erases **all** your addresses, surveys and preferences **on this phone**. Data already synced to the server is **not** deleted server-side — contact the admin for that.

1. Profile → **“Delete my data”**.
2. A confirmation appears: **“Are you sure?”**.
3. Type “DELETE” in the confirmation field.
4. Validate. AsyncStorage is wiped. App returns to onboarding.

12.4 Log out

Profile → **“Log out”**. Your data stays on the phone and on the server. At the next launch, you will be asked to log back in or to continue as guest.

13. FAQ and troubleshooting

Question	Answer
My GPS stays yellow even after 1 minute, what to do?	1) Step outside, away from a building (GPS penetrates metal or concrete roofs poorly). 2) Mov
The district is not auto-detected when I create an address.	Normal if you are offline.. Reverse geocoding (Nominatim) needs Internet. Ente
I created 30 addresses offline, are they lost?	No! They are saved locally and will be automatically synced as soon as you hav
Can anyone see my household data (income, persons...)?	No. Household surveys are strictly private. The server filters by you
My commerce doesn't show up on the users' maps.	1) Verify you chose “Commerce” and not “Household” at the first survey step.<
I can't receive the SMS code at login.	In current demo mode, no real SMS is sent. The code 123456 always works. T
I uninstalled the app by mistake. Is my data lost?	No, as long as you logged in (not guest) and sync happened, your data is saved on the ser
Building photo is very large. How to optimize?	The app automatically compresses photos as JPEG quality 70%. To save more space, choose
Can I use the app abroad (outside Cameroon)?	Yes! The app works worldwide. GPS works everywhere, and reverse ge
The app stopped working, who do I contact?	1) Force close the app and relaunch. 2) Check you have the latest version

14. Appendix

14.1 Glossary

Term	Definition
AsyncStorage	Persistent key/value storage system on the phone, used to keep addresses, surveys ar
Azimuth	Horizontal angle in degrees (0° = North) indicating the entrance door direction.
Household	Survey type representing a dwelling. Data strictly private.
Commerce	Survey type representing a commercial place (shop, workshop, restaurant...). Data pub
Reverse geocoding	Conversion of GPS coordinates to a textual address (district, city). Requires Internet.
Haversine	Mathematical formula to compute the distance between two GPS points on Earth.
Leaflet	Open-source mapping library used to display the interactive map.
MOCK / Demo mode	Current mode where no SMS is sent; the code 123456 always works.
NetInfo	Library that continuously monitors network state to trigger auto-sync.
Nominatim	OpenStreetMap online service that transforms coordinates into district / city.
OTP	One-Time Password — single-use code received via SMS to log in.
OpenStreetMap (OSM)	Collaborative free worldwide map, free equivalent of Google Maps.
QR Code	2D barcode containing your address URL, scannable by any phone.
Sync queue	Local queue of operations to send to the server as soon as network returns.
Twilio	International SMS provider, used to send real OTP codes in production.
Wizard	Step-by-step assistant in several stages (used for address creation and survey).

14.2 Color code recap

Color	Meaning
■ **Primary green (#0D5C3A)**	EFYL brand color, main buttons, verified statuses.
■ **Accent orange (#F97316)**	Important action buttons (New Address), Commerce pins.
■ **Yellow**	Light warnings, “offline” status, medium accuracy.
■ **Red**	Errors, low accuracy, flagged addresses.
■ **Blue**	Neutral information.
■ **Grey**	Secondary text, disabled states, low-accuracy pins.

14.3 Useful shortcuts and gestures

Gesture / Action	Result
Pull down (pull-to-refresh)	Refresh the list on Home or Map.
Long press on an address	Context menu (coming): share, delete, duplicate.
Pinch the map	Zoom in / out.
Drag the orange pin (step 4)	Manually correct the GPS position.
Android Back button	Go back one step in the wizard; on Home, ask confirmation to quit.

14.4 Test credentials (to remove in production)

Identifier	Value
Test phone number	+237 600 000 000 (or any format)
Demo OTP code	123456
Guest mode	Always available, no login required

This guide was automatically generated by the EFYL team. For any question, suggestion or bug report, please reach out to us. Happy mapping! ■■